

EGA GUNTARA

IoT & Embedded Systems Engineer — Firmware, Connectivity & Devices

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SUMMARY

Engineer who builds IoT systems end to end — firmware, electronics, connectivity and deployment — with 5+ years across embedded systems, telecom devices and test engineering. Led IoT projects with Aligness, including a smart greenhouse for the Ministry of Agriculture in service at 5 sites since 2024. Currently a Senior QA Engineer at IDEMIA, validating UICC/eSIM products on live networks and building the team's automation and web tooling. Open to IoT, firmware and embedded systems roles.

EXPERIENCE

IDEMIA — Jakarta, Indonesia

Quality Assurance Engineer 2 (Senior)

Nov 2022 – Present

Quality Assurance Engineer

Mar 2026 – Present

Nov 2022 – Mar 2026

Test Automation & Internal Tooling

- **NVM endurance automation:** Python over AT commands, unattended 24-hour cycles on 5G modems — now used on ~90% of projects.
- **eSIM provisioning tool:** the team's main tool for eUICC projects, cutting stuck-ICCID recovery from six manual steps to a few clicks (Django, PostgreSQL, LDAP, Jenkins, Docker).
- **OTA test platform:** pooled shared test handsets behind a browser (Flask, MQTT, Redis), removing the per-tester local setup.
- **Test-script generator and validator:** converts Comprion trace logs into DLL-ready .pcom scripts, replacing up to 3 business days of manual editing, and corrected a length defect in all 22 GET IDENTITY commands of a ~20,000-command capture.

Validation & Testing

- Validate UICC/eUICC (SIM/eSIM) card OS and applets end to end on live 3G/4G/5G and NB-IoT networks for international mobile operators, down to APDU level with Comprion tooling.
- Run feasibility studies against 3GPP, ISO 7816, ETSI and GSMA SGP specifications; design test cases in Jira/Xray and author the ATD/ATR documents that release a product.
- Test eSIM remote provisioning on live SM-DP/SM-SR platforms (GSMA RSP), and bring up test devices on Quectel, U-blox, Rollingwire and Nordic modules over AT commands.
- Trainer and point of contact for interns over two terms; support other IDEMIA sites (Noida, France, Manila).

Team Aligness (independent engineering team) — IoT Engineer & Project Lead • Freelance

Jan 2023 – Sep 2025

- **Smart Greenhouse, Ministry of Agriculture** — project lead; 5 sites in operation since 2024. Designed the electrical control panels (components, wiring, UPS sizing) and the control path arbitrating automatic vs manual at the power layer, since the commercial fertigation controller exposed no API. Co-developed the ESP32/FreeRTOS firmware, CO2/light sensor nodes on a custom binary protocol, a motorised shading net with hardware interlock, and fleet-wide OTA updates.
- **Flood & Water-Tank Monitoring** — system design and firmware. ESP32 nodes with ultrasonic and float sensors publish over MQTT; an ESP32-P4 wall panel (LVGL, TLS, C on ESP-IDF) decides and commands two tanks, four solenoid valves and the flood pump, which run unattended.
- **Industrial Voltage & Current Monitoring** (power plant client) — system architect: MQTT ingestion from 16 power-sensor lines, impedance calculated with residual compensation against distance-relay protection zones, MongoDB storage, and a Dash dashboard with editable setpoints and alerts, inside a one-second end-to-end budget.
- **BLE Attendance Tracker** (confidential client) — project lead, team of 2, delivered as a working prototype. Raspberry Pi scanner written from scratch in Python: identity from the iBeacon UUID rather than the randomised MAC address, enter/exit from RSSI trend against a threshold, XML reports over FTP every 60 seconds.
- **WiFi Smart Plug** (internal R&D) — ESP32-C3 firmware in C on ESP-IDF, SSR relay control, BL0937 power metering, TLS-secured MQTT, two-stage provisioning, custom 2-layer PCB.

PT Len Industri (Persero) — Embedded System Engineer

Jun 2021 – Nov 2022

- Embedded system engineer on the GCI (ground control intercept) IFF/SSR radar programme, in consortium with PT LAPI ITB, PT RTI and PT Infoglobal for Indonesian Ministry of Defence R&D.
- Designed the Transmit/Receive module on an FPGA board (Vivado) and wrote its firmware in C using Vitis IDE; designed the UDP communication between the TR modules and the beam controller, and the TR module PCB in Eagle.
- Processed radar-captured data with the data processing team and wrote the project specification documentation.

Widya Robotics — Hardware Engineer

Apr 2021 – Jun 2021

- Developed a 3D Concrete Printing (3DCP) machine in hardware and software on Arduino, Raspberry Pi, STM32 and LPC1768 boards: modified Marlin firmware for the printer configuration, wrote a Python G-code generator, and supported the STM32-LPC1768 communication in C/C++.
- Ran functional testing on both the electronics and the software side, and troubleshot faults across the 3DCP system.

- Developed a lift sub-system on Arduino, STM32, Raspberry Pi and relay control in C and Python, and designed and reviewed the PCBs in Eagle.

PROJECTS

- **EspOTAUpdate** (open source) — OTA firmware update library for ESP32 that uses GitHub Pages as the distribution server: push a new version and fielded devices update themselves, with no hosting infrastructure.
- **SIM Tracer** (in progress) — passive T=0 SIM/UICC APDU tracer on STM32, decoding card traffic and streaming it to Wireshark over GSMTAP.
- **Eye-Beacon** (open source, team of 6) — hardware engineer: device hardware for an IoT people-counting system, a Raspberry Pi Zero 2W acting as both RTSP camera and BLE beacon, with a custom PCB add-on and enclosure.

SKILLS

- **Embedded & IoT:** ESP32 (ESP-IDF, FreeRTOS), STM32, Raspberry Pi, embedded C/C++, RTOS, MQTT, BLE/iBeacon, LVGL, OTA updates, UART/SPI/I2C, AT commands, PCB design (Eagle), electrical panel design
- **Telecom & Test:** UICC/eUICC (SIM/eSIM), GSMA RSP (SM-DP/SM-SR), APDU analysis, OTA SMS-PP, 3G/4G/5G, NB-IoT, Quectel/U-blox/Nordic modules, Comprion tooling, ISTQB CTFL, test automation, Jira/Xray
- **Software:** Python, Django, Flask, PostgreSQL, MongoDB, Redis, Docker, Jenkins, LDAP, Linux, Git

EDUCATION & CERTIFICATION

- **Padjadjaran University** — B.Eng. Electrical Engineering, GPA 3.27/4.00 (2016–2020) · **ISTQB CTFL**, 2023